

AI Assisted Coding Assignment- 13

2303A51886 : 30 SAI SPURTHI. B 27.02.2026

Lab 13: Code Refactoring – Improving Legacy Code with AI Suggestions

Lab Objectives:

- Identify code smells and inefficiencies in legacy Python scripts.
- Use AI-assisted coding tools to **refactor** for readability, maintainability, and performance.
- Apply **modern Python best practices** while ensuring output correctness.

Task Description #1 (Refactoring – Removing Code Duplication)

- Task: Use AI to refactor a given Python script that contains multiple repeated code blocks.
- Instructions:
 - Prompt AI to identify duplicate logic and replace it with functions or classes.
 - Ensure the refactored code maintains the same output.
 - Add docstrings to all functions.
- Sample Legacy Code:

```
# Legacy script with repeated logic
print("Area of Rectangle:", 5 * 10)
print("Perimeter of Rectangle:", 2 * (5 + 10))
print("Area of Rectangle:", 7 * 12)
print("Perimeter of Rectangle:", 2 * (7 + 12))
print("Area of Rectangle:", 10 * 15)
print("Perimeter of Rectangle:", 2 * (10 + 15))
```

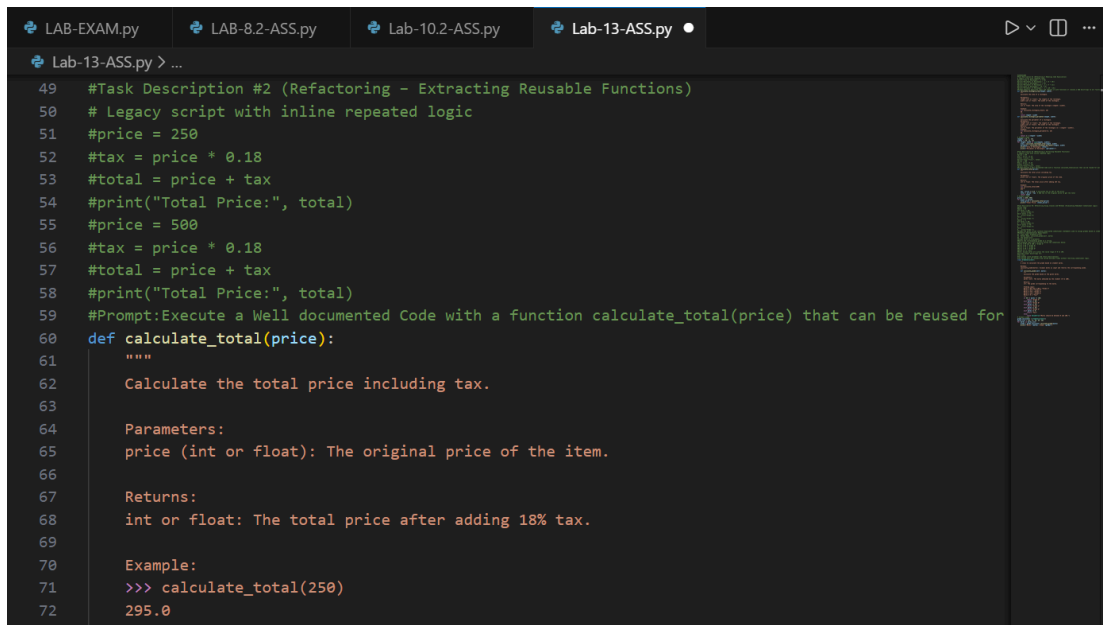
- Expected Output:
 - Refactored code with a reusable function and no duplication.
 - Well documented code

```
Lab-13-ASS.py > ...
2 #Task Description #1 (Refactoring - Removing Code Duplication)
3 # Legacy script with repeated logic
4 #print("Area of Rectangle:", 5 * 10)
5 #print("Perimeter of Rectangle:", 2 * (5 + 10))
6 #print("Area of Rectangle:", 7 * 12)
7 #print("Perimeter of Rectangle:", 2 * (7 + 12))
8 #print("Area of Rectangle:", 10 * 15)
9 #print("Perimeter of Rectangle:", 2 * (10 + 15))
10 #Prompt:Identify duplicate logic and replace it with functions or classes.o Add docstrings to all functions
11 def calculate_rectangle_area(length, width):
12     """
13     Calculate the area of a rectangle.
14
15     Parameters:
16     length (int or float): The length of the rectangle.
17     width (int or float): The width of the rectangle.
18
19     Returns:
20     int or float: The area of the rectangle (length * width).
21
22     Example:
23     >>> calculate_rectangle_area(5, 10)
24     50
25     """
26     return length * width
27 def calculate_rectangle_perimeter(length, width):
28     """
29     Calculate the perimeter of a rectangle.
30
31     Parameters:
32     length (int or float): The length of the rectangle.
33     width (int or float): The width of the rectangle.
34
35     Returns:
36     int or float: The perimeter of the rectangle (2 * (length + width)).
37
38     Example:
39     >>> calculate_rectangle_perimeter(5, 10)
40     30
41     """
```

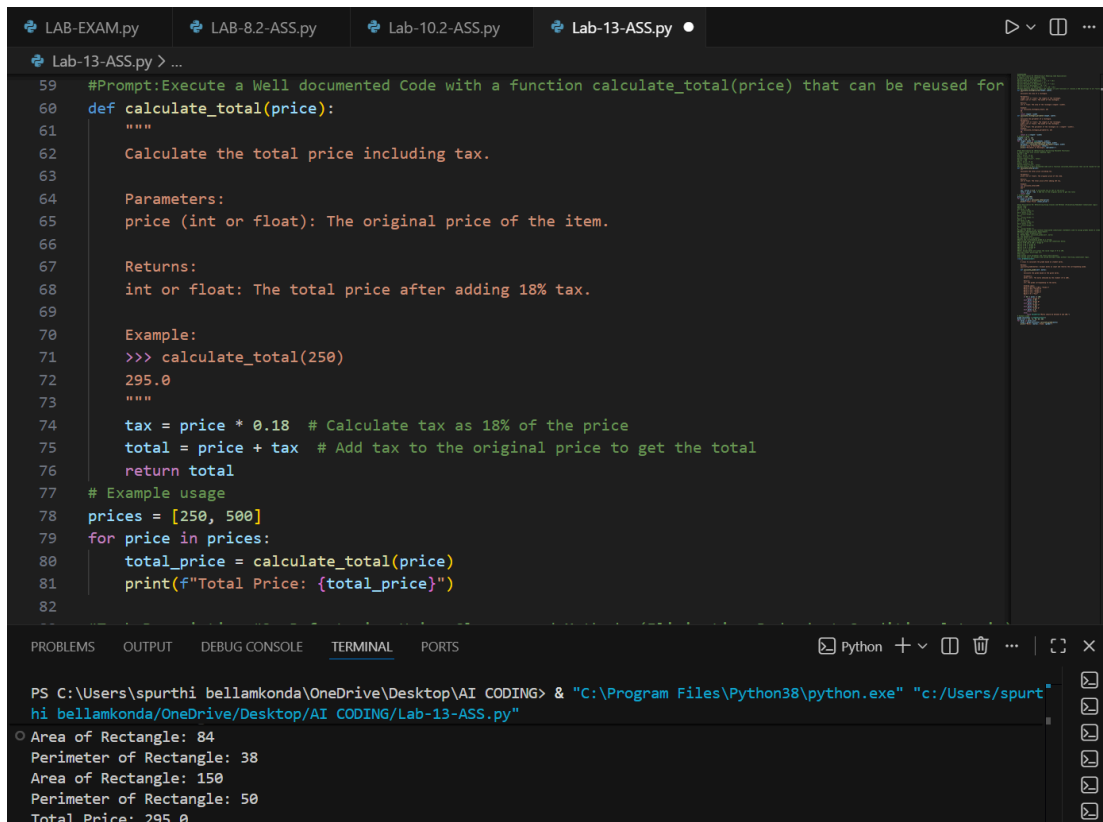


```
print("Total Price:", total)
```

- Expected Output:
 - Code with a function `calculate_total(price)` that can be reused for multiple price inputs.
 - Well documented code



```
49 #Task Description #2 (Refactoring - Extracting Reusable Functions)
50 # Legacy script with inline repeated logic
51 #price = 250
52 #tax = price * 0.18
53 #total = price + tax
54 #print("Total Price:", total)
55 #price = 500
56 #tax = price * 0.18
57 #total = price + tax
58 #print("Total Price:", total)
59 #Prompt:Execute a Well documented Code with a function calculate_total(price) that can be reused for
60 def calculate_total(price):
61     """
62     Calculate the total price including tax.
63
64     Parameters:
65     price (int or float): The original price of the item.
66
67     Returns:
68     int or float: The total price after adding 18% tax.
69
70     Example:
71     >>> calculate_total(250)
72     295.0
```



```
59 #Prompt:Execute a Well documented Code with a function calculate_total(price) that can be reused for
60 def calculate_total(price):
61     """
62     Calculate the total price including tax.
63
64     Parameters:
65     price (int or float): The original price of the item.
66
67     Returns:
68     int or float: The total price after adding 18% tax.
69
70     Example:
71     >>> calculate_total(250)
72     295.0
73     """
74     tax = price * 0.18 # Calculate tax as 18% of the price
75     total = price + tax # Add tax to the original price to get the total
76     return total
77 # Example usage
78 prices = [250, 500]
79 for price in prices:
80     total_price = calculate_total(price)
81     print(f"Total Price: {total_price}")
82
```

```
PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING> & "C:\Program Files\Python38\python.exe" "c:/Users/spurt
hi bellamkonda/OneDrive/Desktop/AI CODING/Lab-13-ASS.py"
Area of Rectangle: 84
Perimeter of Rectangle: 38
Area of Rectangle: 150
Perimeter of Rectangle: 50
Total Price: 295.0
```

Task Description #3: Refactoring Using Classes and Methods (Eliminating Redundant Conditional Logic)

Refactor a Python script that contains repeated if-elif-else grading logic by implementing a structured, object-oriented solution using a class and a method.

Problem Statement

The given script contains duplicated conditional statements used to assign grades based on student marks. This redundancy violates clean code principles and reduces maintainability.

You are required to refactor the script using a class-based design to improve modularity, reusability, and readability while preserving the original grading logic.

Mandatory Implementation Requirements

1. Class Name: GradeCalculator
2. Method Name: calculate_grade(self, marks)
3. The method must:
 - Accept marks as a parameter.
 - Return the corresponding grade as a string.

- The grading logic must strictly follow the conditions below:
- Marks ≥ 90 and $\leq 100 \rightarrow$ "Grade A"
- Marks $\geq 80 \rightarrow$ "Grade B"
- Marks $\geq 70 \rightarrow$ "Grade C"
- Marks $\geq 40 \rightarrow$ "Grade D"
- Marks $\geq 0 \rightarrow$ "Fail"

Note: Assume marks are within the valid range of 0 to 100.

1. Include proper docstrings for:

- The class
- The method (with parameter and return descriptions)

```

317     fib_sequence = [0, 1] # Initialize list with first two values
318     for i in range(2, n):
319         next_value = fib_sequence[-1] + fib_sequence[-2] # Sum of last two numbers
320         fib_sequence.append(next_value) # Append new value to the list
321
322     return fib_sequence
323
324 # Take input from user
325 n = int(input("Enter the number of Fibonacci numbers to generate: "))
326 # Call the function and print the result
327 fibonacci_numbers = generate_fibonacci(n)
328 print(f"Fibonacci sequence up to {n} numbers: {fibonacci_numbers}")
329
330 # Add few test cases to check different values of n
331 print(generate_fibonacci(0)) # Should return []
332 print(generate_fibonacci(1)) # Should return [0]
333 print(generate_fibonacci(5)) # Should return [0, 1, 1, 2, 3]
334 print(generate_fibonacci(10)) # Should return [0, 1, 1, 2, 3, 5, 8, 13, 21, 34]
335
336 ---

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

Enter the number of Fibonacci numbers to generate: 6
Enter the number of Fibonacci numbers to generate: 6
Fibonacci sequence up to 6 numbers: [0, 1, 1, 2, 3, 5]
[]
[0]
[0, 1, 1, 2, 3]
[0, 1, 1, 2, 3, 5, 8, 13, 21, 34]
PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING>

```

2. The method must be reusable and called multiple times without rewriting conditional logic.

- Given code:

```

marks = 85
if marks >= 90:
    print("Grade A")
elif marks >= 75:
    print("Grade B")
else:
    print("Grade C")
marks = 72

```

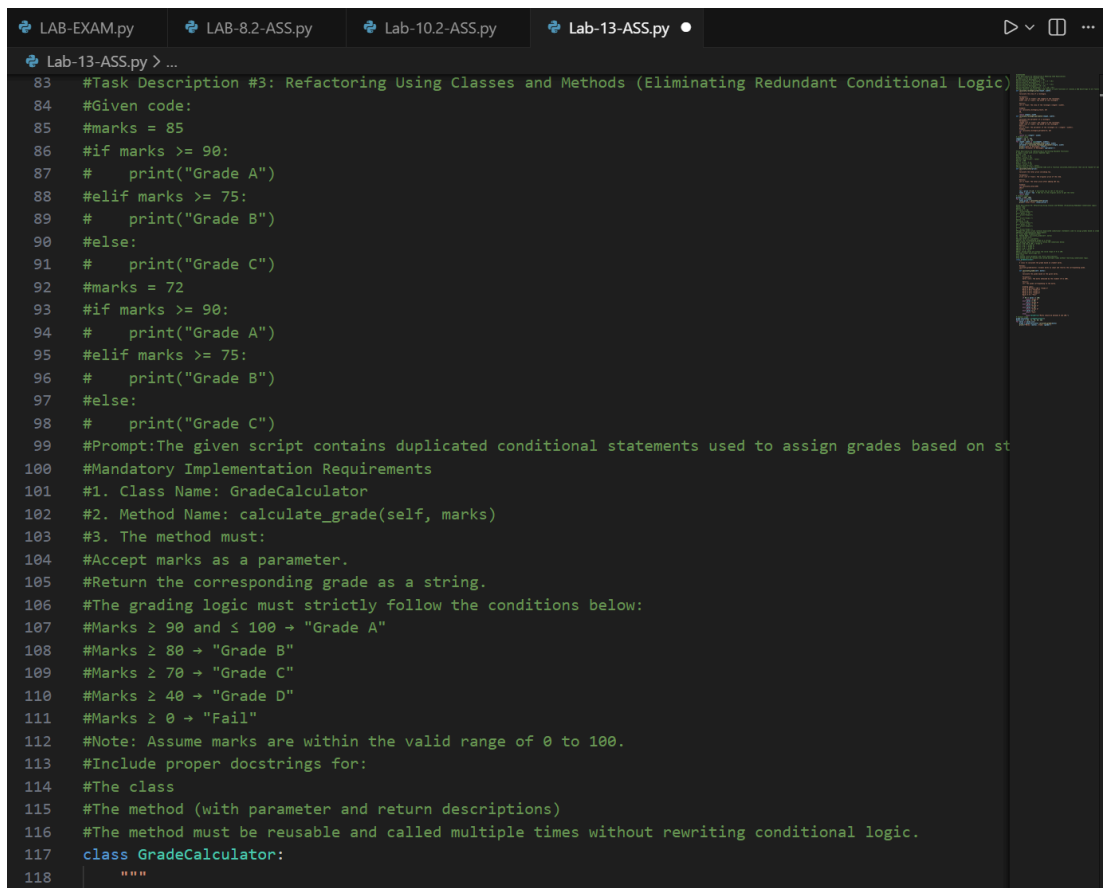
```

if marks >= 90:
    print("Grade A")
elif marks >= 75:
    print("Grade B")
else:
    print("Grade C")

```

Expected Output:

- Define a class named GradeCalculator.
- Implement a method calculate_grade(self, marks) inside the class.
- Create an object of the class.
- Call the method for different student marks.
- Print the returned grade values.



```

LAB-EXAM.py LAB-8.2-ASS.py Lab-10.2-ASS.py Lab-13-ASS.py
Lab-13-ASS.py > ...
83 #Task Description #3: Refactoring Using Classes and Methods (Eliminating Redundant Conditional Logic)
84 #Given code:
85 #marks = 85
86 #if marks >= 90:
87 #    print("Grade A")
88 #elif marks >= 75:
89 #    print("Grade B")
90 #else:
91 #    print("Grade C")
92 #marks = 72
93 #if marks >= 90:
94 #    print("Grade A")
95 #elif marks >= 75:
96 #    print("Grade B")
97 #else:
98 #    print("Grade C")
99 #Prompt:The given script contains duplicated conditional statements used to assign grades based on st
100 #Mandatory Implementation Requirements
101 #1. Class Name: GradeCalculator
102 #2. Method Name: calculate_grade(self, marks)
103 #3. The method must:
104 #Accept marks as a parameter.
105 #Return the corresponding grade as a string.
106 #The grading logic must strictly follow the conditions below:
107 #Marks ≥ 90 and ≤ 100 → "Grade A"
108 #Marks ≥ 80 → "Grade B"
109 #Marks ≥ 70 → "Grade C"
110 #Marks ≥ 40 → "Grade D"
111 #Marks ≥ 0 → "Fail"
112 #Note: Assume marks are within the valid range of 0 to 100.
113 #Include proper docstrings for:
114 #The class
115 #The method (with parameter and return descriptions)
116 #The method must be reusable and called multiple times without rewriting conditional logic.
117 class GradeCalculator:
118     """

```

```
LAB-EXAM.py LAB-8.2-ASS.py Lab-10.2-ASS.py Lab-13-ASS.py
Lab-13-ASS.py > ...
117 class GradeCalculator:
121     Methods:
122     calculate_grade(marks): Accepts marks as input and returns the corresponding grade.
123     """
124     def calculate_grade(self, marks):
125         """
126         Calculate the grade based on the given marks.
127
128         Parameters:
129         marks (int): The marks obtained by the student (0 to 100).
130
131         Returns:
132         str: The grade corresponding to the marks.
133
134         Grading Logic:
135         Marks ≥ 90 and ≤ 100 → "Grade A"
136         Marks ≥ 80 → "Grade B"
137         Marks ≥ 70 → "Grade C"
138         Marks ≥ 40 → "Grade D"
139         Marks ≥ 0 → "Fail"
140         """
141         if 90 <= marks <= 100:
142             return "Grade A"
143         elif marks >= 80:
144             return "Grade B"
145         elif marks >= 70:
146             return "Grade C"
147         elif marks >= 40:
148             return "Grade D"
149         elif marks >= 0:
150             return "Fail"
151         else:
152             raise ValueError("Marks should be between 0 and 100.")
153
154 # Example usage
155 grade_calculator = GradeCalculator()
156 marks_list = [85, 72, 95, 65, 30]
```



```
LAB-EXAM.py LAB-8.2-ASS.py Lab-10.2-ASS.py Lab-13-ASS.py •
Lab-13-ASS.py > ...
161 #Task-4(Code Clarity Improvement)
162 # Function to get input from user and return the number
163 # Function to calculate square of a number and return result
164 # Function to display the result
165 # Main function to call all other functions in order
166 # Call the main function
167 def get_input():
168     """
169     Get a number input from the user.
170
171     Returns:
172     int: The number entered by the user.
173     """
174     return int(input("Enter a number: "))
175 def calculate_square(number):
176     """
177     Calculate the square of a given number.
178     Parameters:
179     number (int): The number to be squared.
180
181     Returns:
182     int: The square of the input number.
183     """
184     return number ** 2
185 def display_result(result):
186     """
187     Display the result to the user.
188     Parameters:
189     result (int): The result to be displayed.
190
191     Returns:
192     None
```

The screenshot shows a VS Code editor with a file explorer on the left containing files: LAB-EXAM.py, LAB-8.2-ASS.py, Lab-10.2-ASS.py, and Lab-13-ASS.py. The active file is Lab-13-ASS.py, which contains the following Python code:

```
185 def display_result(result):
186     """
187     Display the result to the user.
188     Parameters:
189     result (int): The result to be displayed.
190
191     Returns:
192     None
193     """
194     print(f"The square of the number is: {result}")
195
196 def main():
197     """
198     Main function to execute the program flow.
199     This function orchestrates the input, processing, and output of the program.
200     Returns:
201     None
202     """
203     number = get_input() # Get input from the user
204     square = calculate_square(number) # Calculate the square of the number
205     display_result(square) # Display the result to the user
206
207 if __name__ == "__main__":
208     main() # Call the main function to run the program
```

The terminal at the bottom shows the output of the program:

```
Marks: 95, Grade: Grade A
Marks: 65, Grade: Grade D
Marks: 30, Grade: Fail
Marks: 65, Grade: Grade D
Marks: 30, Grade: Fail
Marks: 30, Grade: Fail
Enter a number: 5
The square of the number is: 25
PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING>
```

Task 5 (Refactoring Procedural Code into OOP Design)

- **Task: Use AI to refactor procedural code into a class-based design.** Focus Areas:
 - o Object-Oriented principles
 - o Encapsulation

Legacy Code:

```
salary = 50000
tax = salary * 0.2
net = salary - tax
print(net)
```

Expected Outcome:

- o A class like EmployeeSalaryCalculator with methods and attributes.

Task 6 (Optimizing Search Logic)

- **Task: Refactor inefficient linear searches using appropriate data structures.** Focus Areas: o Time complexity o Data structure choice

Legacy Code:

```
users = ["admin", "guest", "editor", "viewer"]
name = input("Enter username: ")
found = False
for u in users:
    if u == name:
        found = True
print("Access Granted" if found else "Access Denied")
```

Expected Outcome:

- o Use of sets or dictionaries with complexity justification

2. Insert triple quotes under each function and let Copilot complete the docstrings.
3. Generate documentation in the terminal.
4. Export the documentation in HTML format.
5. Open the file in a browser.

Given Code

```
# Library Management System (Unstructured Version)
# This code needs refactoring into a proper module with documentation.

library_db = {}

# Adding first book
title = "Python Basics"
author = "John Doe"
isbn = "101"

if isbn not in library_db:
    library_db[isbn] = {"title": title, "author": author}
    print("Book added successfully.")
else:
    print("Book already exists.")

# Adding second book (duplicate logic)
title = "AI Fundamentals"
author = "Jane Smith"
isbn = "102"

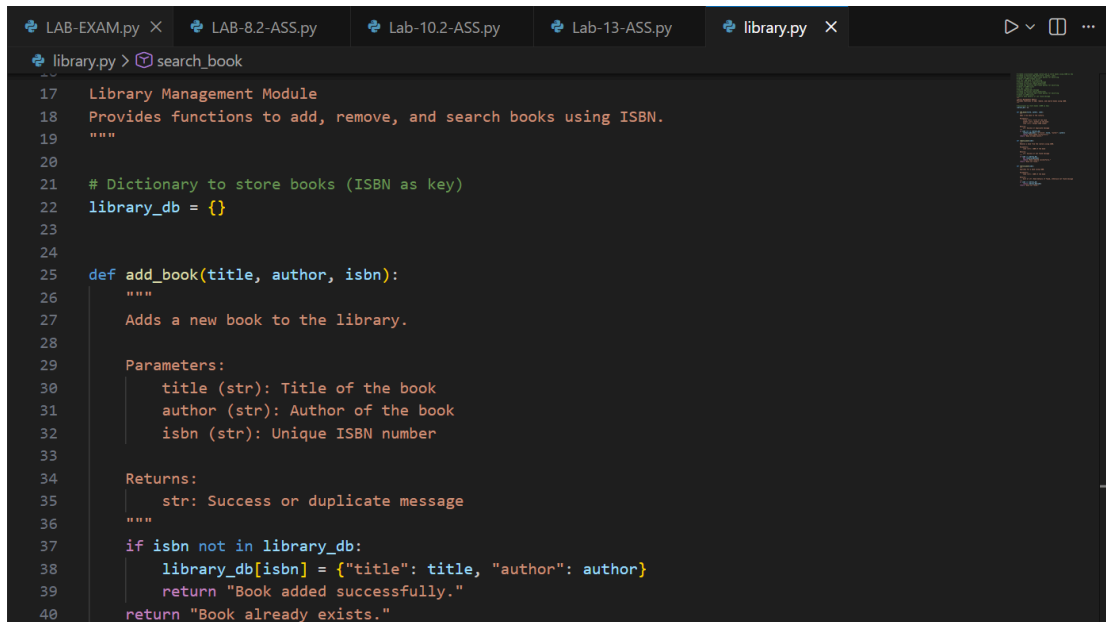
if isbn not in library_db:
    library_db[isbn] = {"title": title, "author": author}
    print("Book added successfully.")
else:
    print("Book already exists.")

# Searching book (repeated logic structure)
isbn = "101"

if isbn in library_db:
    print("Book Found:", library_db[isbn])
else:
    print("Book not found.")

# Removing book (again repeated pattern)
isbn = "101"
```

```
if isbn in library_db:
    del library_db[isbn]
    print("Book removed successfully.")
else:
    print("Book not found.")
# Searching again
isbn = "101"
if isbn in library_db:
    print("Book Found:", library_db[isbn])
else:
    print("Book not found.")
```



```
LAB-EXAM.py X LAB-8.2-ASS.py Lab-10.2-ASS.py Lab-13-ASS.py library.py X
library.py > search_book
17 Library Management Module
18 Provides functions to add, remove, and search books using ISBN.
19 """
20
21 # Dictionary to store books (ISBN as key)
22 library_db = {}
23
24
25 def add_book(title, author, isbn):
26     """
27     Adds a new book to the library.
28
29     Parameters:
30         title (str): Title of the book
31         author (str): Author of the book
32         isbn (str): Unique ISBN number
33
34     Returns:
35         str: Success or duplicate message
36     """
37     if isbn not in library_db:
38         library_db[isbn] = {"title": title, "author": author}
39         return "Book added successfully."
40     return "Book already exists."
```



```
LAB-EXAM.py LAB-8.2-ASS.py Lab-10.2-ASS.py Lab-13-ASS.py library.py X
library.py > search_book
43 def remove_book(isbn):
44     """
45     isbn (str): ISBN of the book
46
47     Returns:
48     str: Success or not found message
49     """
50
51     if isbn in library_db:
52         del library_db[isbn]
53         return "Book removed successfully."
54     return "Book not found."
55
56
57
58
59 def search_book(isbn):
60     """
61     Searches for a book using ISBN.
62
63     Parameters:
64     isbn (str): ISBN of the book
65
66     Returns:
67     dict or str: Book details if found, otherwise not found message
68     """
69
70     if isbn in library_db:
71         return library_db[isbn]
72     return "Book not found."
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```

The screenshot shows the VS Code editor with the 'library.html' file open. The file contains HTML code for a library management interface. The code includes a header section with a title 'Python: module library' and a meta tag for content type. The main content area features a table with a heading 'Library Management Module' and a section 'Functions'. The 'Functions' section lists three functions: 'add_book', 'remove_book', and 'search_book', each with its parameters and return values. The rendered output in the browser shows the same content, with the title 'library' and the description 'Library Management Module' and 'Provides functions to add, remove, and search books using ISBN.' The 'Functions' section is highlighted in orange and lists the same three functions with their details.

Task 8– Fibonacci Generator

Write a program to generate Fibonacci series up to n.

The initial code has:

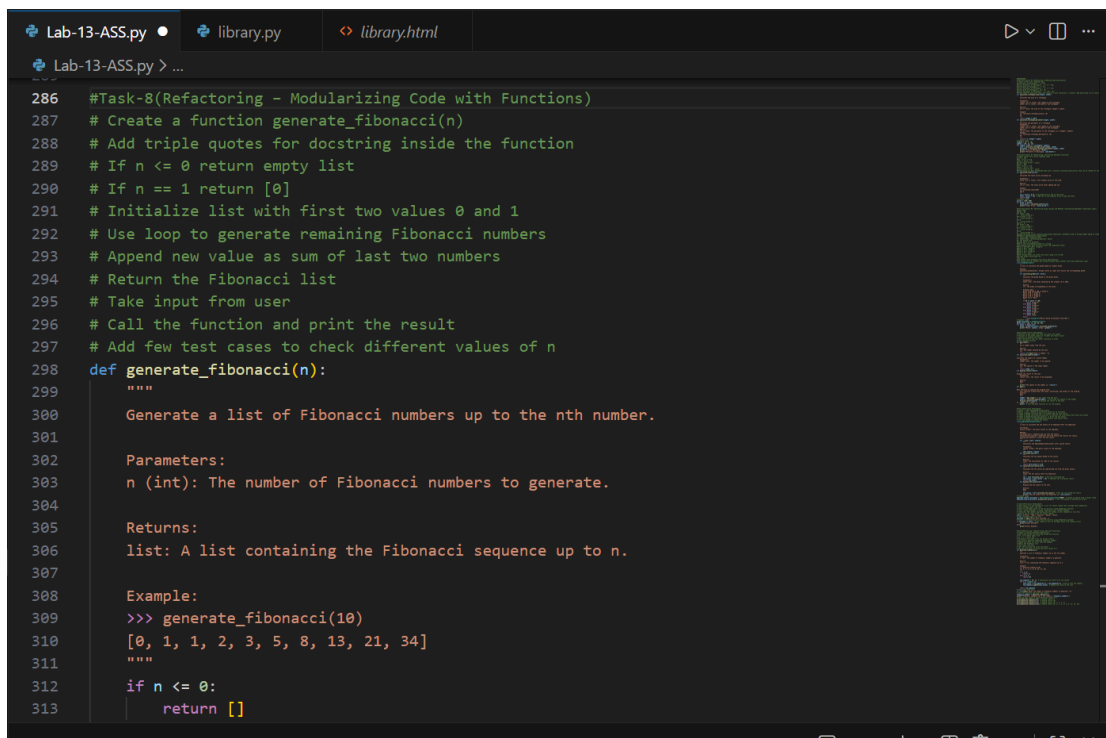
- Global variables.
- Inefficient loop.
- No functions or modularity.

Task for Students:

- Refactor into a clean reusable function (generate_fibonacci).
- Add docstrings and test cases.
- Compare AI-refactored vs original.

Bad Code Version:

```
# fibonacci bad version
n=int(input("Enter limit: "))
a=0
b=1
print(a)
print(b)
for i in range(2,n):
    c=a+b
    print(c)
    a=b
    b=c
```



The screenshot shows a code editor with a dark theme. The active file is 'Lab-13-ASS.py'. The code is as follows:

```
286 #Task-8(Refactoring - Modularizing Code with Functions)
287 # Create a function generate_fibonacci(n)
288 # Add triple quotes for docstring inside the function
289 # If n <= 0 return empty list
290 # If n == 1 return [0]
291 # Initialize list with first two values 0 and 1
292 # Use loop to generate remaining Fibonacci numbers
293 # Append new value as sum of last two numbers
294 # Return the Fibonacci list
295 # Take input from user
296 # Call the function and print the result
297 # Add few test cases to check different values of n
298 def generate_fibonacci(n):
299     """
300     Generate a list of Fibonacci numbers up to the nth number.
301
302     Parameters:
303     n (int): The number of Fibonacci numbers to generate.
304
305     Returns:
306     list: A list containing the Fibonacci sequence up to n.
307
308     Example:
309     >>> generate_fibonacci(10)
310     [0, 1, 1, 2, 3, 5, 8, 13, 21, 34]
311     """
312     if n <= 0:
313         return []
```

```
Lab-13-ASS.py • library.py <> library.html
Lab-13-ASS.py > ...
298 def generate_fibonacci(n):
317     fib_sequence = [0, 1] # Initialize list with first two values
318     for i in range(2, n):
319         next_value = fib_sequence[-1] + fib_sequence[-2] # Sum of last two numbers
320         fib_sequence.append(next_value) # Append new value to the list
321
322     return fib_sequence
323
324 # Take input from user
325 n = int(input("Enter the number of Fibonacci numbers to generate: "))
326 # Call the function and print the result
327 fibonacci_numbers = generate_fibonacci(n)
328 print(f"Fibonacci sequence up to {n} numbers: {fibonacci_numbers}")
329
330 # Add few test cases to check different values of n
331 print(generate_fibonacci(0)) # Should return []
332 print(generate_fibonacci(1)) # Should return [0]
333 print(generate_fibonacci(5)) # Should return [0, 1, 1, 2, 3]
334 print(generate_fibonacci(10)) # Should return [0, 1, 1, 2, 3, 5, 8, 13, 21, 34]
---
```

```
PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING> & "C:\Program Files\Python38\python.exe" "c:/Users/spurti bellamkonda/OneDrive/Desktop/AI CODING/Lab-13-ASS.py"
Access Denied
Enter the number of Fibonacci numbers to generate: 6
Fibonacci sequence up to 6 numbers: [0, 1, 1, 2, 3, 5]
[]
[0]
[0, 1, 1, 2, 3]
Enter the number of Fibonacci numbers to generate: 6
Fibonacci sequence up to 6 numbers: [0, 1, 1, 2, 3, 5]
[]
[0]
Enter the number of Fibonacci numbers to generate: 6
Fibonacci sequence up to 6 numbers: [0, 1, 1, 2, 3, 5]
Enter the number of Fibonacci numbers to generate: 6
Enter the number of Fibonacci numbers to generate: 6
```

Task 9 – Twin Primes Checker

Twin primes are pairs of primes that differ by 2 (e.g., 11 and 13, 17 and 19).

The initial code has:

- Inefficient prime checking.
- No functions.
- Hardcoded inputs.

Task for Students:

- Refactor into `is_prime(n)` and `is_twin_prime(p1, p2)`.
- Add docstrings and optimize.
- Generate a list of twin primes in a given range using AI.

Bad Code Version:

```
# twin primes bad version
```

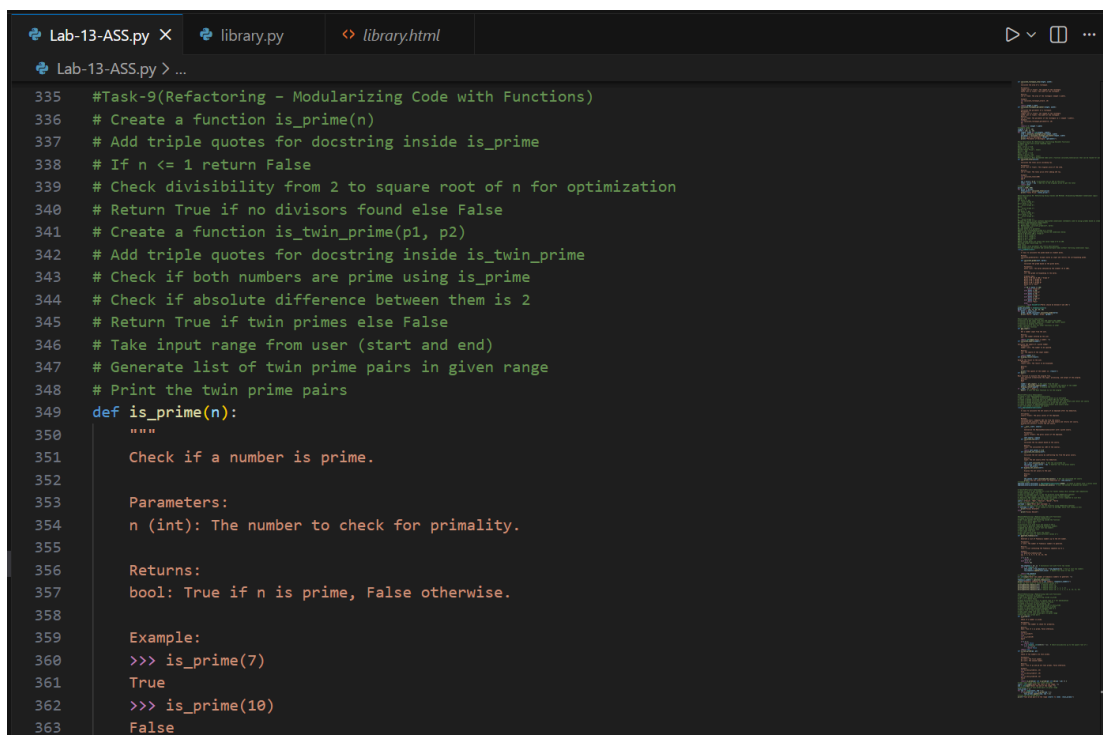
```
a=11
```

```
b=13
```

```

fa=0
for i in range(2,a):
    if a%i==0:
        fa=1
fb=0
for i in range(2,b):
    if b%i==0:
        fb=1
if fa==0 and fb==0 and abs(a-b)==2:
    print("Twin Primes")
else:
    print("Not Twin Primes")

```



```

Lab-13-ASS.py X library.py library.html
Lab-13-ASS.py > ...
335 #Task-9(Refactoring - Modularizing Code with Functions)
336 # Create a function is_prime(n)
337 # Add triple quotes for docstring inside is_prime
338 # If n <= 1 return False
339 # Check divisibility from 2 to square root of n for optimization
340 # Return True if no divisors found else False
341 # Create a function is_twin_prime(p1, p2)
342 # Add triple quotes for docstring inside is_twin_prime
343 # Check if both numbers are prime using is_prime
344 # Check if absolute difference between them is 2
345 # Return True if twin primes else False
346 # Take input range from user (start and end)
347 # Generate list of twin prime pairs in given range
348 # Print the twin prime pairs
349 def is_prime(n):
350     """
351     Check if a number is prime.
352
353     Parameters:
354     n (int): The number to check for primality.
355
356     Returns:
357     bool: True if n is prime, False otherwise.
358
359     Example:
360     >>> is_prime(7)
361     True
362     >>> is_prime(10)
363     False

```


The current program reads a year from the user and prints the corresponding Chinese Zodiac sign. However, the implementation contains repetitive conditional logic, lacks modular design, and does not follow clean coding principles.

Your task is to refactor the code to improve readability, maintainability, and structure.

Chinese Zodiac Cycle (Repeats Every 12 Years)

1. Rat
2. Ox
3. Tiger
4. Rabbit
5. Dragon
6. Snake
7. Horse
8. Goat (Sheep)
9. Monkey
10. Rooster
11. Dog
12. Pig

Chinese Zodiac Program (Unstructured Version)

This code needs refactoring.

```
year = int(input("Enter a year: "))
if year % 12 == 0:
    print("Monkey")
elif year % 12 == 1:
    print("Rooster")
elif year % 12 == 2:
    print("Dog")
elif year % 12 == 3:
    print("Pig")
elif year % 12 == 4:
    print("Rat")
elif year % 12 == 5:
    print("Ox")
elif year % 12 == 6:
    print("Tiger")
```

```
elif year % 12 == 7:
    print("Rabbit")
elif year % 12 == 8:
    print("Dragon")
elif year % 12 == 9:
    print("Snake")
elif year % 12 == 10:
    print("Horse")
elif year % 12 == 11:
    print("Goat")
```

You must:

1. Create a reusable function: `get_zodiac(year)`
 2. Replace the if-elif chain with a cleaner structure (e.g., list or dictionary).
 3. Add proper docstrings.
 4. Separate input handling from logic.
 5. Improve readability and maintainability.
 6. Ensure output remains correct.
-

Refactor the given poorly structured Python script into a clean, modular, and reusable implementation.

A Harshad (Niven) number is a number that is divisible by the sum of its digits. **For example:**

- $18 \rightarrow 1 + 8 = 9 \rightarrow 18 \div 9 = 2$  (Harshad Number)
- $19 \rightarrow 1 + 9 = 10 \rightarrow 19 \div 10 \neq \text{integer}$  (Not Harshad)

Problem Statement

The current implementation:

- Mixes logic and input handling
- Uses redundant variables
- Does not use reusable functions properly
- Returns print statements instead of boolean values
- Lacks documentation

You must refactor the code to follow clean coding principles.

Harshad Number Checker (Unstructured Version)

```
num = int(input("Enter a number: "))
```

```
temp = num
```

```
sum_digits = 0
```

```
while temp > 0:
```

```
    digit = temp % 10
```

```
    sum_digits = sum_digits + digit
```

```
    temp = temp // 10
```

```
if sum_digits != 0:
```

```
    if num % sum_digits == 0:
```

```
        print("True")
```

```
    else:
```

```
        print("False")
```

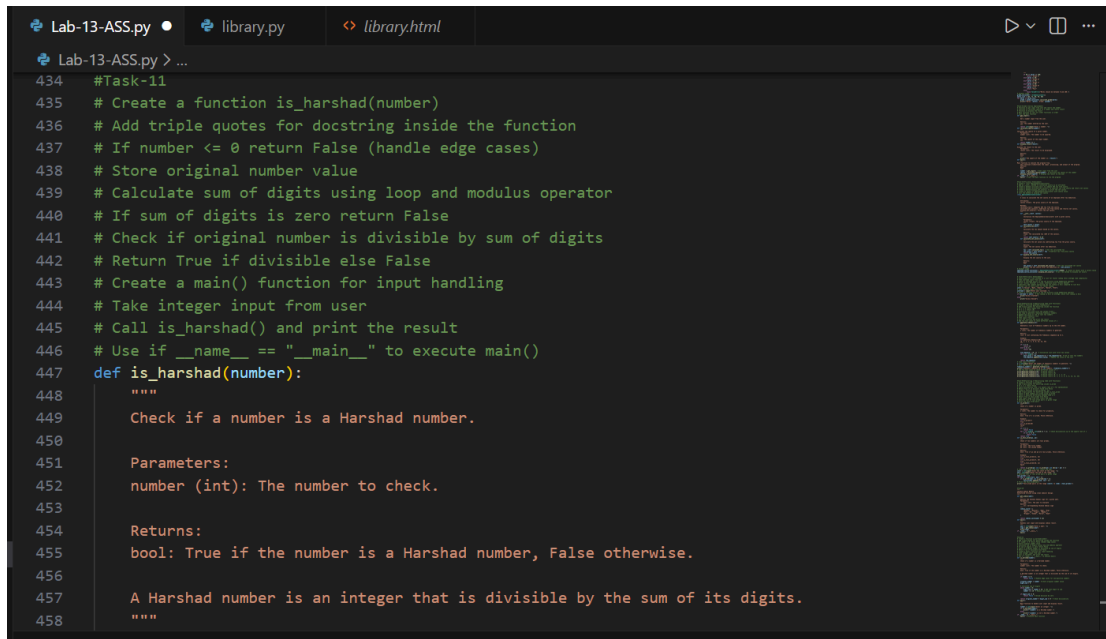
```
else:
```

```
    print("False")
```

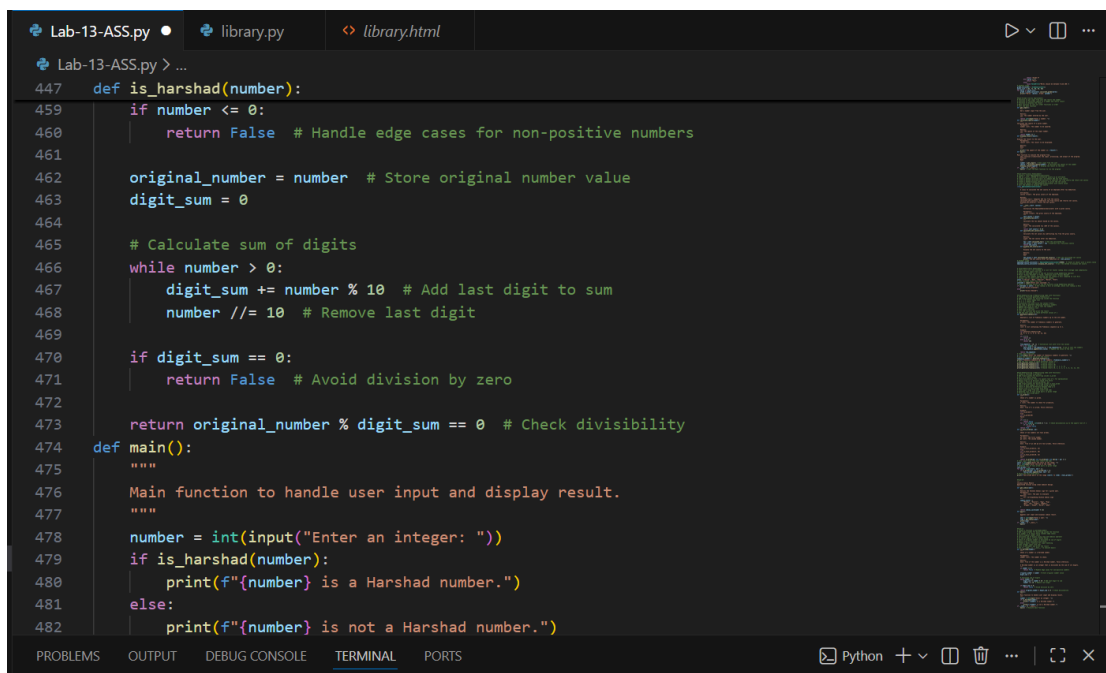
You must:

1. Create a reusable function: `is_harshad(number)`
2. The function must:
 - o Accept an integer parameter.
 - o Return True if the number is divisible by the sum of its digits.
 - o Return False otherwise.
3. Separate user input from core logic.

4. Add proper docstrings.
 5. Improve readability and maintainability.
 6. Ensure the program handles edge cases (e.g., 0, negative numbers).
-



```
Lab-13-ASS.py • library.py library.html
Lab-13-ASS.py > ...
434 #Task-11
435 # Create a function is_harshad(number)
436 # Add triple quotes for docstring inside the function
437 # If number <= 0 return False (handle edge cases)
438 # Store original number value
439 # Calculate sum of digits using loop and modulus operator
440 # If sum of digits is zero return False
441 # Check if original number is divisible by sum of digits
442 # Return True if divisible else False
443 # Create a main() function for input handling
444 # Take integer input from user
445 # Call is_harshad() and print the result
446 # Use if __name__ == "__main__" to execute main()
447 def is_harshad(number):
448     """
449     Check if a number is a Harshad number.
450
451     Parameters:
452     number (int): The number to check.
453
454     Returns:
455     bool: True if the number is a Harshad number, False otherwise.
456
457     A Harshad number is an integer that is divisible by the sum of its digits.
458     """
```



```
Lab-13-ASS.py • library.py library.html
Lab-13-ASS.py > ...
447 def is_harshad(number):
448     """
449     Check if a number is a Harshad number.
450
451     Parameters:
452     number (int): The number to check.
453
454     Returns:
455     bool: True if the number is a Harshad number, False otherwise.
456
457     A Harshad number is an integer that is divisible by the sum of its digits.
458     """
459     if number <= 0:
460         return False # Handle edge cases for non-positive numbers
461
462     original_number = number # Store original number value
463     digit_sum = 0
464
465     # Calculate sum of digits
466     while number > 0:
467         digit_sum += number % 10 # Add last digit to sum
468         number //= 10 # Remove last digit
469
470     if digit_sum == 0:
471         return False # Avoid division by zero
472
473     return original_number % digit_sum == 0 # Check divisibility
474 def main():
475     """
476     Main function to handle user input and display result.
477     """
478     number = int(input("Enter an integer: "))
479     if is_harshad(number):
480         print(f"{number} is a Harshad number.")
481     else:
482         print(f"{number} is not a Harshad number.")
```

```

477     """
478     number = int(input("Enter an integer: "))
479     if is_harshad(number):
480         print(f"{number} is a Harshad number.")
481     else:
482         print(f"{number} is not a Harshad number.")
483 if __name__ == "__main__":
484     main() # Execute main function
485
486

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Python + v [Icons] [Close]

```

C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING> python.exe C:/Users/spurthi bellamkonda/OneDrive/Desktop/AI CODING/12. Refactoring the Factorial Trailing Zeros Program.py
[0, 1, 1, 2, 3]
[0, 1, 1, 2, 3, 5, 8, 13, 21, 34]
Enter the start of the range: 1
Enter the end of the range: 20
Twin prime pairs in the range 1 to 20: [(3, 5), (5, 7), (11, 13), (17, 19)]
Enter a year: 2024
Dragon
Enter an integer: 18
18 is a Harshad number.
PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING>

```

Task 12 – Refactoring the Factorial Trailing Zeros Program

Refactor the given poorly structured Python script into a clean, modular, and efficient implementation.

The program calculates the number of trailing zeros in $n!$ (factorial of n).

Problem Statement

The current implementation:

- Calculates the full factorial (inefficient for large n)
- Mixes input handling with business logic
- Uses print statements instead of return values
- Lacks modular structure and documentation

You must refactor the code to improve efficiency, readability, and maintainability.

Factorial Trailing Zeros (Unstructured Version)

```
n = int(input("Enter a number: "))
```

```
fact = 1
```

```
i = 1
```

```
while i <= n:
```

```
    fact = fact * i
```

```
    i = i + 1
```

```
count = 0
```

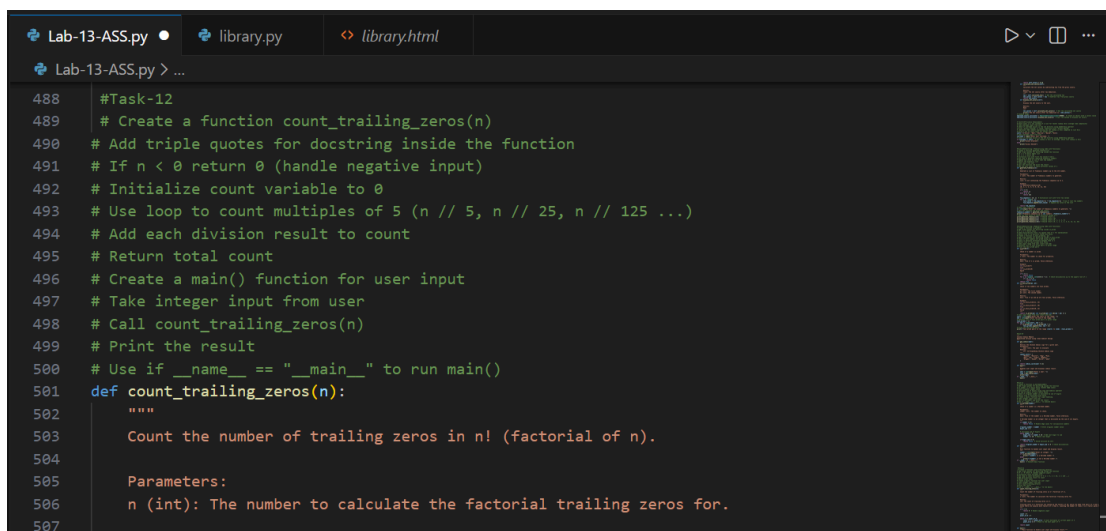
```
while fact % 10 == 0:
```

```
    count = count + 1
```

```
fact = fact // 10
print("Trailing zeros:", count)
```

You must:

1. Create a reusable function: `count_trailing_zeros(n)`
2. The function must:
 - o Accept a non-negative integer `n`.
 - o Return the number of trailing zeros in `n!`.
3. Do NOT compute the full factorial.
4. Use an optimized mathematical approach (count multiples of 5).
5. Add proper docstrings.
6. Separate user interaction from core logic.
7. Handle edge cases (e.g., negative numbers, zero).



```
Lab-13-ASS.py • library.py library.html
Lab-13-ASS.py > ...
488 #Task-12
489 # Create a function count_trailing_zeros(n)
490 # Add triple quotes for docstring inside the function
491 # If n < 0 return 0 (handle negative input)
492 # Initialize count variable to 0
493 # Use loop to count multiples of 5 (n // 5, n // 25, n // 125 ...)
494 # Add each division result to count
495 # Return total count
496 # Create a main() function for user input
497 # Take integer input from user
498 # Call count_trailing_zeros(n)
499 # Print the result
500 # Use if __name__ == "__main__" to run main()
501 def count_trailing_zeros(n):
502     """
503     Count the number of trailing zeros in n! (factorial of n).
504
505     Parameters:
506     n (int): The number to calculate the factorial trailing zeros for.
507
```

The screenshot shows a code editor with a file named 'Lab-13-ASS.py'. The code defines a function `count_trailing_zeros(n)` that counts the number of trailing zeros in $n!$ by repeatedly dividing n by 5. It also includes a `main()` function that prompts the user for a non-negative integer and prints the result. The terminal output shows the execution of the script, including prompts for a range, twin prime pairs, a year, and the final result for $5!$.

```

501 def count_trailing_zeros(n):
520     while n >= power_of_5:
521         count += n // power_of_5 # Count multiples of current power of 5
522         power_of_5 *= 5 # Move to the next power of 5
523
524     return count
525
526 def main():
527     """Main function to handle user input and display result."""
528
529     n = int(input("Enter a non-negative integer: "))
530     zeros_count = count_trailing_zeros(n)
531     print(f"The number of trailing zeros in {n}! is: {zeros_count}")
532
533 if __name__ == "__main__":
534     main() # Execute main function

```

```

PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING> & "C:\Program Files\Python38\python.exe" "c:/Users/spurthi bellamkonda/OneDrive/Desktop/AI CODING/Lab-13-ASS.py"
[0]
[0, 1, 1, 2, 3]
[0, 1, 1, 2, 3, 5, 8, 13, 21, 34]
Enter the start of the range: 1
Enter the end of the range: 20
Twin prime pairs in the range 1 to 20: [(3, 5), (5, 7), (11, 13), (17, 19)]
Enter a year: 2024
Dragon
Enter an integer: 7
7 is a Harshad number.
Enter a non-negative integer: 5
The number of trailing zeros in 5! is: 1
PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING>

```

Task 13 (Collatz Sequence Generator – Test Case Design)

- Function: Generate Collatz sequence until reaching 1.
- Test Cases to Design:
 - Normal: $6 \rightarrow [6, 3, 10, 5, 16, 8, 4, 2, 1]$
 - Edge: $1 \rightarrow [1]$
 - Negative: -5
 - Large: 27 (well-known long sequence)
- Requirement: Validate correctness with pytest.

Explanation:

We need to write a function that:

- Takes an integer n as input.
- Generates the Collatz sequence (also called the $3n+1$ sequence).
- The rules are:
 - o If n is even $\rightarrow$ next = $n / 2$.
 - o If n is odd $\rightarrow$ next = $3n + 1$.
- Repeat until we reach 1.

- Return the full sequence as a list.

Example

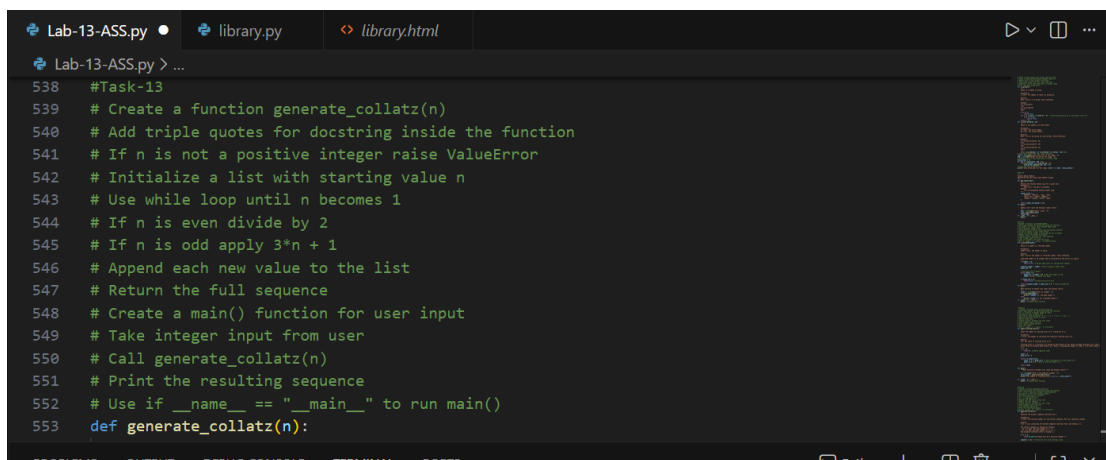
Input: 6

Steps:

- 6 (even $\rightarrow 6/2 = 3$)
- 3 (odd $\rightarrow 3*3+1 = 10$)
- 10 (even $\rightarrow 10/2 = 5$)
- 5 (odd $\rightarrow 3*5+1 = 16$)
- 16 (even $\rightarrow 16/2 = 8$)
- 8 (even $\rightarrow 8/2 = 4$)
- 4 (even $\rightarrow 4/2 = 2$)
- 2 (even $\rightarrow 2/2 = 1$)

Output:

[6, 3, 10, 5, 16, 8, 4, 2, 1]



The screenshot shows a code editor with a dark theme. The top bar has three tabs: 'Lab-13-ASS.py', 'library.py', and 'library.html'. The active tab is 'Lab-13-ASS.py'. The code is a Python script for the Collatz conjecture. It includes a function 'generate_collatz(n)' and a 'main()' function. The code is as follows:

```
538 #Task-13
539 # Create a function generate_collatz(n)
540 # Add triple quotes for docstring inside the function
541 # If n is not a positive integer raise ValueError
542 # Initialize a list with starting value n
543 # Use while loop until n becomes 1
544 # If n is even divide by 2
545 # If n is odd apply 3*n + 1
546 # Append each new value to the list
547 # Return the full sequence
548 # Create a main() function for user input
549 # Take integer input from user
550 # Call generate_collatz(n)
551 # Print the resulting sequence
552 # Use if __name__ == "__main__" to run main()
553 def generate_collatz(n):
```

```
Lab-13-ASS.py • library.py <> library.html
Lab-13-ASS.py > ...
553 def generate_collatz(n):
567     """
568     if n <= 0:
569         raise ValueError("Input must be a positive integer.")
570
571     sequence = [n] # Initialize list with starting value
572
573     while n != 1:
574         if n % 2 == 0: # If n is even
575             n //= 2
576         else: # If n is odd
577             n = 3 * n + 1
578         sequence.append(n) # Append each new value to the list
579
580     return sequence
581
582 def main():
583     """Main function to handle user input and display the Collatz sequence."""
584
585     n = int(input("Enter a positive integer: "))
586     try:
587         collatz_sequence = generate_collatz(n)
588         print(f"Collatz sequence starting from {n}: {collatz_sequence}")
589     except ValueError as e:
590         print(e)
591
592 if __name__ == "__main__":
593     main() # Execute main function

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python + - []

Enter an integer: 8
8 is a Harshad number.
Enter a non-negative integer: 6
The number of trailing zeros in 6! is: 1
Enter a positive integer: 5
Collatz sequence starting from 5: [5, 16, 8, 4, 2, 1]
```

Task 14 (Lucas Number Sequence – Test Case Design)

- Function: Generate Lucas sequence up to n terms. (Starts with 2,1, then $F_n = F_{n-1} + F_{n-2}$) Test Cases to Design:
- Normal: 5 → [2, 1, 3, 4, 7]
- Edge: 1 → [2]
- Negative: -5 → Error
- Large: 10 (last element = 76). Requirement: Validate correctness with pytest.


```
Lab-13-ASS.py • library.py library.html
Lab-13-ASS.py > ...
594 #Task-14
595 # Create a function generate_lucas(n)
596 # Add triple quotes for docstring inside the function
597 # If n is not a positive integer raise ValueError
598 # If n == 1 return [2]
599 # Initialize list with first two values [2, 1]
600 # Use loop to generate remaining Lucas numbers
601 # Append sum of last two numbers to the list
602 # Return the full Lucas sequence
603 # Add main block for manual testing
604 # Take integer input from user
605 # Call generate_lucas function
606 # Print the result
607 def generate_lucas(n):
608     """
609     Generate the Lucas sequence up to the nth number.
610
611     Parameters:
612     n (int): The number of Lucas numbers to generate. Must be a positive integer.
613
614     Returns:
615     list: A list containing the Lucas sequence up to n.
616
617     The Lucas sequence is defined as follows:
618     - L(0) = 2
619     - L(1) = 1
```

```
Lab-13-ASS.py • library.py library.html
Lab-13-ASS.py > ...
607 def generate_lucas(n):
623     raise ValueError("Input must be a positive integer.")
624
625     if n == 1:
626         return [2] # Return the first Lucas number
627
628     lucas_sequence = [2, 1] # Initialize list with first two values
629
630     for i in range(2, n):
631         next_value = lucas_sequence[-1] + lucas_sequence[-2] # Sum of last two numbers
632         lucas_sequence.append(next_value) # Append new value to the list
633
634     return lucas_sequence
635 def main():
636     """Main function to handle user input and display the Lucas sequence."""
637
638     n = int(input("Enter the number of Lucas numbers to generate: "))
639     try:
640         lucas_numbers = generate_lucas(n)
641         print(f"Lucas sequence up to {n} numbers: {lucas_numbers}")
642     except ValueError as e:
643         print(e)
644 if __name__ == "__main__":
645     main() # Execute main function
```

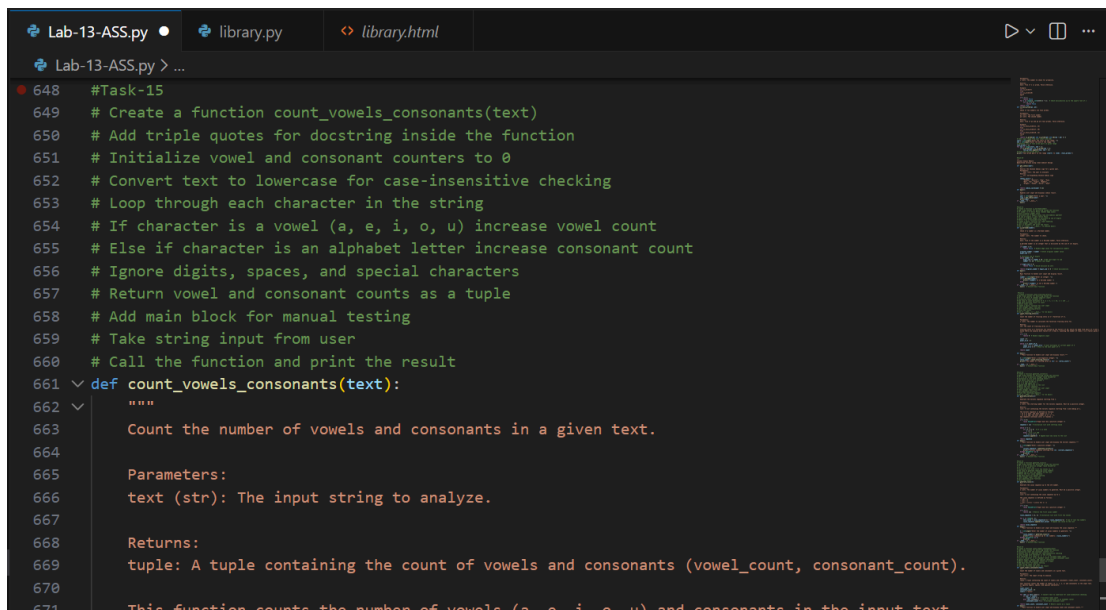
7 is a Harshad number.
Enter a non-negative integer: 5
The number of trailing zeros in 5! is: 1
Enter a positive integer: 3
Collatz sequence starting from 3: [3, 10, 5, 16, 8, 4, 2, 1]
Enter the number of Lucas numbers to generate: -5
Input must be a positive integer.
PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING>

Task 15 (Vowel & Consonant Counter – Test Case Design)

- Function: Count vowels and consonants in string. Test Cases to Design:
- Normal: "hello" → (2,3)
- Edge: "" → (0,0)
- Only vowels: "aeiou" → (5,0)

Large: Long text

- Requirement: Validate correctness with pytest.



```
Lab-13-ASS.py • library.py library.html
Lab-13-ASS.py > ...
648 #Task-15
649 # Create a function count_vowels_consonants(text)
650 # Add triple quotes for docstring inside the function
651 # Initialize vowel and consonant counters to 0
652 # Convert text to lowercase for case-insensitive checking
653 # Loop through each character in the string
654 # If character is a vowel (a, e, i, o, u) increase vowel count
655 # Else if character is an alphabet letter increase consonant count
656 # Ignore digits, spaces, and special characters
657 # Return vowel and consonant counts as a tuple
658 # Add main block for manual testing
659 # Take string input from user
660 # Call the function and print the result
661 def count_vowels_consonants(text):
662     """
663     Count the number of vowels and consonants in a given text.
664
665     Parameters:
666     text (str): The input string to analyze.
667
668     Returns:
669     tuple: A tuple containing the count of vowels and consonants (vowel_count, consonant_count).
670
671     This function counts the number of vowels (a, e, i, o, u) and consonants in the input text
```

```
Lab-13-ASS.py • library.py library.html
Lab-13-ASS.py > ...
661 def count_vowels_consonants(text):
674     vowel_count = 0
675     consonant_count = 0
676     vowels = "aeiou"
677
678     for char in text.lower(): # Convert text to lowercase for case-insensitive checking
679         if char in vowels:
680             vowel_count += 1 # Increase vowel count
681         elif char.isalpha(): # Check if character is an alphabet letter
682             consonant_count += 1 # Increase consonant count
683
684     return vowel_count, consonant_count # Return counts as a tuple
685
686 def main():
687     """Main function to handle user input and display vowel and consonant counts."""
688
689     text = input("Enter a string: ")
690     vowels, consonants = count_vowels_consonants(text)
691     print(f"Number of vowels: {vowels}")
692     print(f"Number of consonants: {consonants}")
693
694 if __name__ == "__main__":
695     main() # Execute main function

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Python + v [Icons]

```
5 is a Harshad number.
Enter a non-negative integer: 6
The number of trailing zeros in 6! is: 1
Enter a positive integer: -7
Input must be a positive integer.
Enter the number of Lucas numbers to generate: -6
Input must be a positive integer.
Enter a string: hello
Number of vowels: 2
Number of consonants: 3

```